

Learning-Based Optimal Right-Turn Coordination System for Connected and Automated Vehicles at Intersections

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Abstract—At urban intersections, traffic flow is often disrupted by uncooperative driving behavior, particularly at four-way signalized intersections lacking a dedicated turn lane or protected signal phase, leading to congestion, increased fuel consumption, and emissions. In left-hand traffic systems (e.g., Japan), right-turning vehicles must yield to oncoming traffic during the green phase. When no safe gap is available, the turning vehicle stops, potentially blocking following vehicles and causing queue spillback over multiple cycles. This paper proposes a Vehicle-to-Everything (V2X) coordination framework with two connected components. First, a Bayesian Gaussian Process Regression (GPR) model estimates free-flow arrival times of approaching vehicles at the stop line. These estimates feed into a slot search optimizer that identifies the least-delay gap in oncoming traffic. Second, a finite-horizon Quadratic Programming Model Predictive Controller (QP-MPC) controls each Connected and Automated Vehicle (CAV) to precisely meet the assigned crossing time while respecting acceleration and gap safety constraints. A factorial ablation over both components, evaluated under identical stochastic traffic conditions, shows that the fuel benefit comes primarily from pairing QP-MPC control with RSU-assigned crossing times. The proposed system achieves the lowest fuel consumption across all demand levels, a 37% reduction over uncoordinated driving, at the cost of a gap-waiting time longer than IDM-coordinated driving, though still precisely tracked rather than oscillatory.

I. INTRODUCTION

Right-turning vehicles at signalized intersections in left-hand traffic environments (e.g., Japan, the United Kingdom, and Australia) must cross the path of oncoming straight-through traffic during the shared green phase. In the absence of a dedicated turn lane or a protected turn signal, the turning vehicle must wait at the stop line until a safe gap naturally appears in the opposing stream. During this waiting period, all following vehicles on the same approach are blocked, forming a queue that may extend over multiple vehicles and persist into the subsequent red phase, thereby triggering spillback congestion [1]. Prior research has demonstrated that such turn-induced delays significantly increase fuel consumption and emissions [2].

Connected and Automated Vehicles (CAVs) equipped with V2X communication offer a viable solution for efficient traffic flow at intersections. A Road Side Unit (RSU) at the intersection aggregates vehicle states broadcast via Vehicle-to-Infrastructure (V2I) links and solves a centralized gap assignment problem at the start of each green phase, coordinating the arrival times of all approaching vehicles to create

a safe gap for turning vehicles. For signalized intersections, eco-approach speed advisory systems leverage V2I signal phase and timing (SPaT) data to reduce idling [2], [3]. Decentralized energy optimal control at signal-free intersections was proposed in [4]. Cooperative merging and trajectory optimization under connectivity have been studied in [5], [6]. Prior work on RSU-based right-turn coordination for left-hand traffic has shown improvement in fuel consumption and speed compared to the uncoordinated baselines; however, these methods rely on the traditional human driver model, specifically the Intelligent Driver Model (IDM), for both vehicle control and arrival time prediction [5]. This can limit coordination accuracy due to stochastic perturbations, parameter mismatch, and the difficulty of tracking assigned crossing times.

In this paper, we propose a coordination framework for CAVS at signalized intersections using V2X. This paper makes the following contributions: (1) A Bayesian GPR estimator that learns the vehicle arrival time mapping offline, eliminating dependence on vehicle dynamics parameter calibration and providing always finite estimates for the RSU slot search optimizer. (2) A QP-MPC formulation is developed for CAV longitudinal control. It uses a finite prediction horizon which enforces hard safety constraints, and a principled cost structure that enables precise tracking of RSU-assigned crossing time. (3) A factorial ablation across five configurations that isolates the controller and estimator contributions independently under identical stochastic traffic conditions, ensuring that any performance differences are attributable solely to the control and estimation methods.

II. PROBLEM STATEMENT

We consider a four-way signalized intersection with single-lane approaches under left-hand traffic, as illustrated in Fig. 1. An Eastbound vehicle that wishes to turn right must cross Westbound straight through traffic during the shared green phase. No dedicated right-turn lane or protected turn signal phase is provided, which is a common configuration at low-volume urban intersections in Japan [7]. The turning vehicle must wait at the stop line until a natural gap appears in the Westbound stream. During this wait, all following Eastbound vehicles are blocked, forming a queue that can grow to span multiple vehicles and potentially force all of them to wait through an additional red cycle.

The RSU at the intersection receives state information from all approaching vehicles via V2I communication and solves a centralized optimization at the start of each green phase to assign coordinated crossing times. Since the quality

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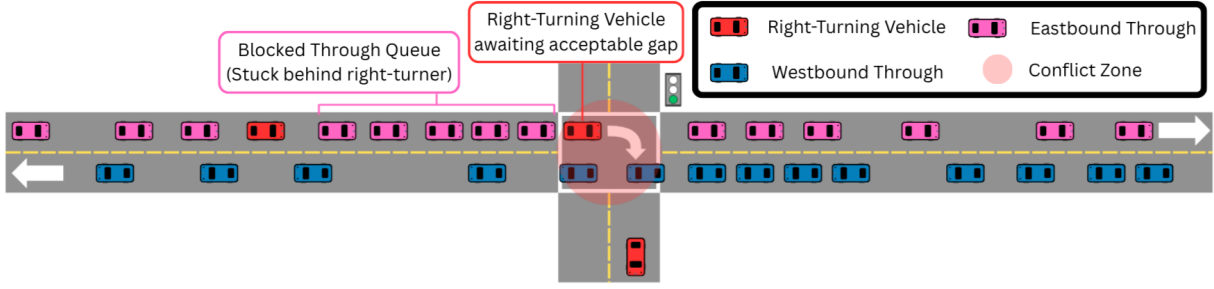


Fig. 1: Right-turn problem at a signalized Intersection. The Eastbound turning vehicle (red) waits at the stop line for a safer gap in the Eastbound (blue). Following Eastbound vehicles (pink) form a blocked queue.

of gap-slot allocation depends directly on these estimates, the main challenge is accurately estimating each vehicle's free-flow arrival time at the stop line.

III. SYSTEM MODEL

A. Intersection and Signal

We model a single-lane, four-way intersection under left-hand traffic, with the intersection center at the origin. Eastbound and Westbound vehicles travel along the horizontal axis; Northbound and Southbound along the vertical axis. In each approach, the stop line is placed shortly before the intersection center. Vehicles spawn at a fixed distance from the center and are removed when clearing the intersection zone. In the reported experiments, only the East-West corridor is active, so right-turn conflicts occur exclusively between Eastbound turning vehicles and Westbound straight-through traffic. A fixed-time signal controller governs the intersection, with each phase consisting of a green interval followed by a short yellow clearance interval and an all-red interval shared by both phases, in compliance with the Japanese traffic engineering guidelines [7]. The signal state is computed deterministically from the cycle clock; all three experimental configurations share identical signal timing.

B. Vehicle Dynamics

All vehicles share the same kinematic model. The state is updated using second-order Euler integration at each discrete time step Δt , with acceleration $a(t)$ as the control input:

$$x_n(t + \Delta t) = x_n(t) + v_n(t)\Delta t + \frac{1}{2}a_n(t)\Delta t^2, \quad (1)$$

$$v_n(t + \Delta t) = \max(0, v_n(t) + a_n(t)\Delta t), \quad (2)$$

where x , v , and a denote the position, velocity, and acceleration, respectively. The kinematic model defined in (1)–(2) is common to all configurations; however, the calculation of $a(t)$ at each time step varies depending on the specific control strategy, as described below.

a) *Traditional Human Driving Model*: In uncoordinated and baseline V2V configurations, acceleration $a(t)$ is computed using the IDM [8]

$$a = a_{\max} \left[1 - \left(\frac{v}{v_d} \right)^4 - \left(\frac{s^*(v, \Delta v)}{s} \right)^2 \right], \quad (3)$$

where $s^*(v, \Delta v) = S_0 + vT_{hw} + v\Delta v/(2\sqrt{a_{\max}b})$ is the desired gap, S_0 is the minimum standstill spacing, T_{hw} is the desired time headway (sampled per vehicle from a uniform distribution), $a_{\max}$ is the maximum acceleration, b is the comfortable deceleration, and Δv is the relative speed to

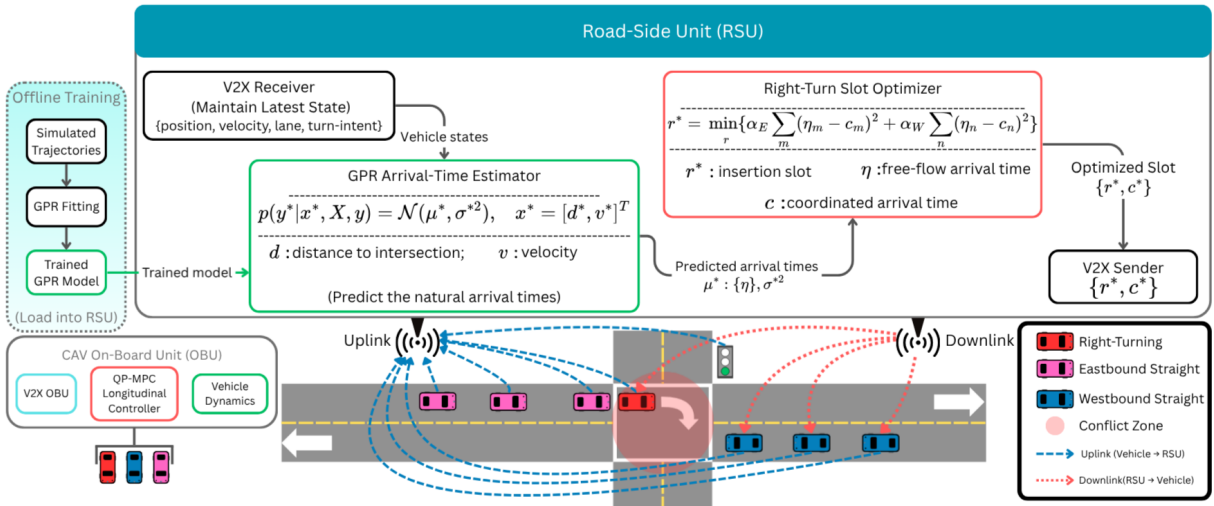


Fig. 2: Overview of the proposed coordination system for CAVs at the intersection.

the lead vehicle. To model realistic human driving behavior, a small zero-mean Gaussian noise term $\mathcal{N}(0, \sigma)$ is applied to the acceleration output, capturing stochastic fluctuations in driver response. This perturbation is applied only to human driving-based configurations. Despite producing a realistic traffic flow, IDM remains a reactive model; it lacks a prediction horizon and cannot optimize control objectives or accurately track externally assigned crossing times. These limitations motivate the need for a principled automation model in the coordination context.

b) Coordinated Control Model: Various models have been proposed for coordinated CAV control, including rule-based, optimization-based, and learning-based methods. Among these, MPC is a good choice for coordination because it can optimize over a finite horizon while enforcing safety constraints and tracking RSU-assigned crossing times. In this work, a Quadratic Programming MPC (QP-MPC) formulation is adopted for its computational efficiency and suitability for real-time implementation, given its convex structure with quadratic cost and linear constraints [9].

State-space prediction model: The vehicle state is defined based on the kinematic model in (1)–(2), where the state vector is given by $z(k) = [s(k) \ v(k)]^T$, representing position and velocity, respectively. From (1) and (2), the system can be written in state-space form as $z(k+1) = Az(k) + Bu(k)$, where $A = \begin{bmatrix} 1 & \Delta t; 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} \frac{1}{2}\Delta t^2 & \Delta t \end{bmatrix}^T$. Over a prediction horizon N , the predicted velocity and position are: $\mathbf{v}_{\text{pred}} = A_v + S_v \mathbf{U}$, $\mathbf{s}_{\text{pred}} = A_s + S_s \mathbf{U}$, where $\mathbf{U} = [u(0) \ u(1) \ \dots \ u(N-1)]^T$, A_v and A_s represent the free evolution of velocity and position, S_v and S_s represent the lower-triangular matrices that capture the influence of the control inputs.

Cost function: The objective penalizes velocity tracking error and control effort:

$$\begin{aligned} \min \quad & \sum_{k=0}^{N-1} [w_v (v_k - v_{d,k})^2 + w_u u_k^2], \\ \text{s.t.} \quad & u_{\min} \leq u_k \leq u_{\max}, \\ & 0 \leq v_k \leq v_{\max}, \\ & s_k \leq s_{\text{lead},k} - s_0 - \Delta s_k, \end{aligned} \quad (4)$$

where $s_{\text{lead},k}$ represents the predicted lead vehicle position under a constant velocity assumption, and Δs_k is a safety margin at the initial steps of the prediction horizon.

IV. RIGHT-TURN COORDINATION OPTIMIZATION

Each vehicle within the communication range broadcasts its state (position, velocity, direction, and turning intent) to the RSU via the V2I uplink at each time step. The communication is assumed to be ideal, meaning no packet loss and negligible latency. The received data are stored in a long-term buffer for offline analysis and GPR training, while the short-term buffer keeps only the most recent messages. The short-term buffer is cleared at the start of each green phase to provide fresh state estimates for the optimizer. Smoothed velocity and estimated acceleration are derived from the short-term buffer.

A. Problem Formulation

At the start of each EW green phase, the RSU collects V2X state data from all vehicles within communication range. Let $a_1, \dots, a_M$ and $b_1, \dots, b_N$ denote the Eastbound and Westbound vehicles, ordered by their estimated arrival times. Let η_m and η_n denote the corresponding free-flow arrival times at the stop line. A Westbound right-turning vehicle b_q must cross the Eastbound stream, while an Eastbound right-turning vehicle a_q must cross the Westbound stream. Based on this, the RSU determines the optimal insertion slot for each right-turning vehicle in the opposing stream and computes coordinated arrival times c_m and c_n that minimize the total squared delay. The following formula jointly minimizes the delay for both right-turning movements by finding the optimal insertion slot r^* in the opposing stream.

$$\begin{aligned} J(r) &= \alpha_E \sum_m (\eta_m - c_m)^2 + \alpha_W \sum_n (\eta_n - c_n)^2, \\ r^* &= \arg \min_{r \in \{1, \dots, M+1\}} J(r), \end{aligned} \quad (5)$$

where α_E and α_W are direction weights (set to equal priority in this work). Vehicles ahead of the coordination point are not delayed, i.e., $c_m = \eta_m$ for $m < r$ and $c_n = \eta_n$ for $n < q$. The search over r is exhaustive. Since there are at most 15 vehicles per phase, each evaluation has complexity $\mathcal{O}(M + N)$, and the total computation remains within the V2X feedback budget.

B. Coordinated Arrival Times

Without loss of generality, we focus on the turning vehicle b_q crossing the Eastbound stream, with a safety gap Δ_{safe} and minimum headway $h_{\min}$. The coordinated times are assigned as follows. For clarity, we focus only on b_q while the case of a_q crossing the Westbound stream is symmetric. The vehicle b_q must cross after a_{r-1} clears and before a_r arrives:

$$c_{a_r} = \max(c_{b_q} + \Delta_{\text{safe}}, \eta_{a_r}). \quad (7)$$

$$c_{b_q} = \max(c_{a_{r-1}} + \Delta_{\text{safe}}, \eta_{b_q}). \quad (8)$$

Subsequent vehicles maintain a minimum headway as:

$$c_{a_m} = \max(c_{a_{m-1}} + h_{\min}, \eta_{a_m}), \quad m = r+1, \dots, M$$

$$c_{b_n} = \max(c_{b_{n-1}} + h_{\min}, \eta_{b_n}), \quad n = q+1, \dots, N$$

Two special cases are handled without invoking the optimizer: when multiple right-turners exist on the same approach, slots are assigned iteratively; when both approaches simultaneously have a right-turning vehicle, no conflict exists, and both receive an immediate proceed signal.

V. BAYESIAN GPR ARRIVAL-TIME PREDICTOR

1) Motivation and Data Collection: The slot-search optimizer requires a free-flow arrival time estimate η_i for each vehicle. Following the learning methodology in [10], [11], we learn this mapping offline from recorded data, rather than relying on vehicle-dynamics-based forward simulation. Training data is collected from the uncoordinated human-driving simulation. Each sample pair's input features $\mathbf{x}_i =$

$[d_i, v_i]^\top$, where d_i is the distance to the stop line and v_i is the current velocity, with output $y_i = \eta_i$, the realized free-flow arrival time.

2) *GPR Model*: A zero-mean Gaussian Process (GP) with squared-exponential (RBF) kernel [12] is used:

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2l^2}\right). \quad (9)$$

The posterior predictive distribution at a test point $\mathbf{x}^* = [d^*, v^*]^\top$ is Gaussian with mean and variance:

$$\mu^*(\mathbf{x}^*) = k(\mathbf{x}^*, X) [K + \sigma_n^2 I]^{-1} \mathbf{y}, \quad (10)$$

$$\sigma^{*2}(\mathbf{x}^*) = k(\mathbf{x}^*, \mathbf{x}^*) - k(\mathbf{x}^*, X) [K + \sigma_n^2 I]^{-1} k(X, \mathbf{x}^*), \quad (11)$$

where $K \in \mathbb{R}^{n \times n}$ is the kernel matrix, $X \in \mathbb{R}^{n \times 2}$ is the training input matrix, and $\mathbf{y} = [y_1, \dots, y_n]^\top$ is the training output vector. Hyperparameters $\{\ell, \sigma_f^2, \sigma_n^2\}$ are learned by maximizing the log marginal likelihood. The posterior mean $\mu^*(\mathbf{x}^*)$ is used as the free-flow arrival time estimate η_i , while the posterior variance $\sigma^{*2}(\mathbf{x}^*)$ provides a measure of predictive uncertainty for potential extensions, such as triggering re-optimization under high uncertainty. The GPR model is pre-trained offline and deployed for inference.

VI. EXPERIMENTAL SETUP

A. Simulation Configurations

The proposed system differs from the reference coordination scheme along two independent axes: the longitudinal controller (IDM vs. QP-MPC) and the RSU arrival-time estimator (IDM forward simulation vs. GPR). To attribute performance to each axis separately, we evaluate the full factorial in addition to two uncoordinated baselines, with all configurations sharing an identical pre-generated stochastic traffic schedule, signal timing, and intersection geometry:

- **C1 (human driving)**: IDM with Gaussian noise perturbation; no V2X; right turns by local gap acceptance.
- **C2 (IDM + RSU_{IDM})**: IDM-based control; RSU slot search driven by IDM forward simulation. Reference coordination scheme.

- **C2G (IDM + RSU_{GPR})**: identical to C2 except the RSU estimator is replaced by GPR, isolating the contribution of the proposed estimator under IDM control.
- **C3I (QP-MPC + RSU_{IDM})**: identical to the proposed system except the RSU retains IDM forward simulation, isolating the contribution of the proposed controller.
- **C3 (proposed, QP-MPC + RSU_{GPR})**: both components active.

B. Simulation Environment

Each configuration is simulated over a 300 s horizon with $\Delta t = 0.5$ s, under three Poisson arrival-rate stages (light/medium/heavy: 0.25/0.40/0.55 veh/s) applied identically across configurations. The QP-MPC controller uses control-effort weight $w_u = 9.0$ and bounds $u \in [-7, 2]$ m/s²; the RSU enforces a safety margin $\Delta_{\text{safe}} = 4.0$ s and minimum headway $h_{\min} = 2.0$ s. Remaining IDM and signal-timing parameters follow standard values consistent with [8] and Japanese traffic engineering guidelines [7].

VII. RESULTS AND DISCUSSION

A. GPR Prediction Accuracy

The GPR-based arrival-time predictor is evaluated in two stages. Model diagnostics across 20 trials (Fig. 3) show GPR generalizes well on held-out data (RMSE= 5.40 s, MAE= 1.44 s), with near-zero residuals and a monotonic (d, v) surface. On $n = 636$ matched samples, GPR achieves the lowest RMSE (9.22 s vs. 12.08 s for IDM forward simulation and 87.97 s for the naive baseline d/v) and always returns a finite estimate, unlike IDM, which fails in 18.1% of cases; GPR's slight positive bias (+0.71 s vs. IDM's -5.42 s) further yields safer, more conservative slot assignments.

The model is trained offline on 12,206 samples from 20 independent human-driving trials (distinct seeds, 1,500 s each): each pre-crossing observation yields a training pair $\mathbf{x}_i = [d_i, v_i]^\top \rightarrow y_i = \eta_i$. A random 2,000-sample subset handles hyperparameter optimization; the trained model is then fixed and deployed unchanged across all RSU-GPR configurations.

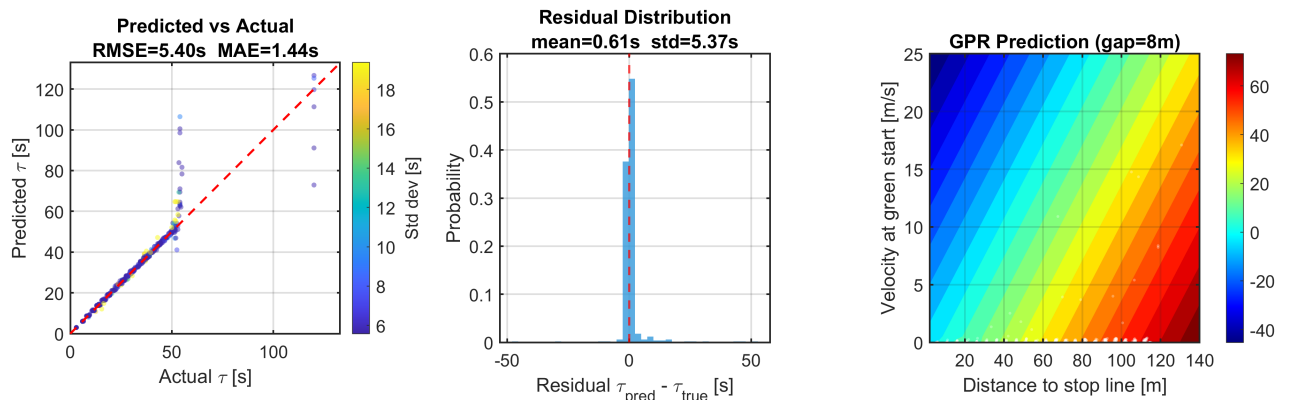


Fig. 3: GPR arrival-time diagnostics on held-out test data. Left: predicted vs. actual crossing time (RMSE= 5.40 s, MAE= 1.44 s). Center: residual distribution (mean= +0.61 s). Right: predicted arrival-time surface over (d, v) space.

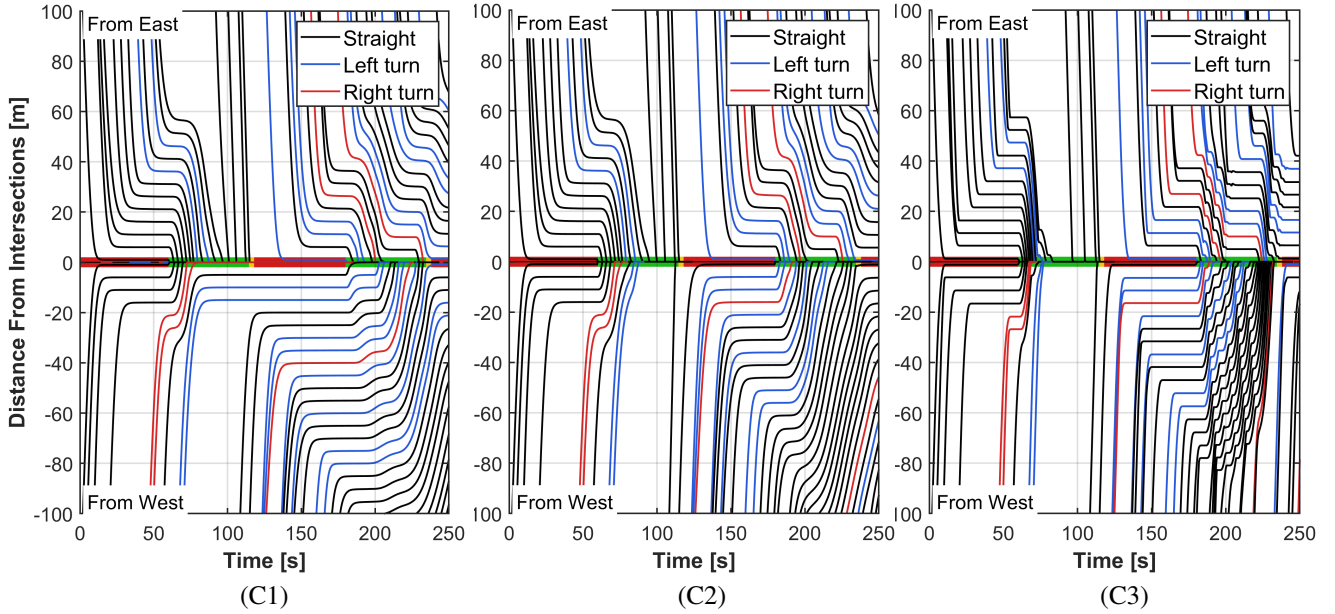


Fig. 4: Trajectories of vehicles passing the intersection for (C1) traditional human driving, (C2) connected and coordinated human driving, and (C3) the proposed MPC with GPR-RSU coordination.

B. Intersection Performance: Trajectory Analysis

Fig. 4 shows representative East–West trajectories for C1, C2, and C3 (C2G and C3I are omitted as visually near-identical to C2 and C3, per Table I). Black, blue, and red lines denote straight, left-turning, and right-turning vehicles, respectively; a right-turner’s dwell time near the stop line reflects its gap-waiting idle time.

Traditional human driving (C1): This configuration shows clear stop-and-wait behavior, with right-turning vehicles showing delays of 20–45s near the stop line. Variations in IDM dynamics and time headways (T_{hw}) create irregular gaps, which are often too small during the green phase. As a result, vehicles are pushed into the next red cycles, reducing intersection throughput.

Coordinated and connected human driving (C2): The RSU reduces gap-waiting time by creating coordinated slots in the Eastbound stream. However, stochastic variations in IDM cause vehicles to miss their assigned crossing times. This leads to visible oscillations in the approach trajectories, where vehicles slow down toward the assigned slot and then slightly speed up again, resulting in repeated deceleration–acceleration cycles. As a result, fuel consumption increases even though the gap-waiting time is shorter.

Proposed system (C3): The proposed system addresses this mismatch. The QP-MPC controller tracks the assigned crossing times, while the GPR estimator provides consistent, always-finite arrival-time predictions. This leads to a single smooth deceleration rather than repeated restart events, at the cost of a longer, more conservative dwell time at the stop line, and more reliable clearance within each green phase.

C. Fuel Consumption Analysis

Fig. 5 summarizes fuel consumption and idle-time characteristics for the representative configurations C1, C2, and

C3; Table I reports the full five-configuration comparison.

Completion is 65% (11/17) under uncoordinated human driving, versus 71% (5/7) for every RSU-coordinated configuration alike, since the RSU defers turns lacking a safe slot rather than admitting marginal gaps the way local acceptance does. All four RSU-coordinated configurations complete the identical five vehicles, so the comparisons in Table I are paired.

C2 shortens gap-waiting but increases fuel use for straight-going vehicles, since IDM repeatedly misses its assigned slot and must slow down and speed up again. C3 avoids this: it has the lowest fuel use of any configuration, even though vehicles wait longer at the stop line. QP-MPC decelerates once, smoothly, to a slightly more conservative crossing time instead of stopping and restarting.

Separating the two changes shows the fuel savings come almost entirely from the controller (C2→C3I), not the estimator: swapping IDM forward simulation for GPR gives the exact same result under both IDM control (C2→C2G) and QP-MPC control (C3I→C3). This is expected: once the schedule and conditions are shared, small differences between estimators stop changing which slot the RSU assigns. The real benefit of GPR is not a different result, but the same result at far lower computational cost: IDM forward simulation steps through up to 120 simulated timesteps per vehicle, while GPR needs only one closed-form evaluation. Two auxiliary V2V-only baselines confirm the separate pattern: uncoordinated IDM (62.10 mL) and uncoordinated QP-MPC (63.48 mL) perform about the same, showing QP-MPC only helps once the RSU gives it a target to track.

RT [mL] and Str [mL] denote the average fuel consumption per completed right-turning and straight-going vehicle, respectively. Gap [s] is the average gap-waiting time of

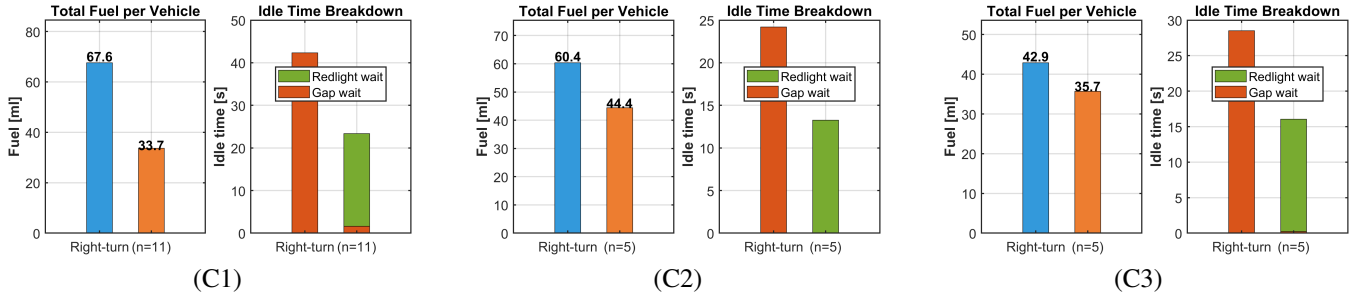


Fig. 5: Per-vehicle fuel consumption and idle-time comparison for (C1) traditional human driving, (C2) IDM with RSU, and (C3) the proposed MPC with GPR-RSU.

TABLE I: Ablation over controller and estimator (identical traffic schedule). RT/Str: mean fuel per completed right-turning / straight-going vehicle. Gap: mean gap-waiting time. Compl.: right turns completed within the 300s horizon / injected.

Metric	C1	C2	C2G	C3I	C3
Controller	IDM*	IDM	IDM	MPC	MPC
RSU estimator	—	IDM	GPR	IDM	GPR
RT [mL]	67.59	60.35	60.35	42.90	42.90
Str [mL]	34.45	45.53	45.53	36.38	36.38
Gap [s]	42.32	24.20	24.20	28.50	28.50
Compl. [-]	11/17	5/7	5/7	5/7	5/7

*IDM with Gaussian noise (human-driving model).

completed right-turning vehicles at the stop line. Compl. reports the number of right-turning vehicles that completed their turn within the 300 s simulation horizon out of the number injected with right-turn intent.

VIII. CONCLUSION

This paper presented a V2X-based right-turn coordination system for CAVs at a Japanese-style signalized intersection: a Bayesian GPR arrival-time estimator trained offline, and a QP-MPC controller for precise crossing-time tracking. A factorial ablation shows the fuel benefit comes primarily from pairing QP-MPC with RSU-assigned crossing times, since QP-MPC alone, without a target to track, performs comparably to IDM alone. GPR’s role is independent of the controller: it guarantees a finite estimate is always available, replacing IDM’s 18.1% failure rate with 0%. The combined system reduces right-turning fuel by 37% over uncoordinated driving, at the cost of a gap-waiting time longer than IDM-coordinated driving, though still precisely tracked rather than oscillatory.

Future work will focus on experimental validation under real-world communication latency and packet loss, and extension to multi-intersection coordination and mixed traffic conditions.

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